Act 3.3

Trayecto 1

exponencial

[0 8]

$f(x) = e^{0.2t}$

$x(t) = t$ $\qquad \dot{x}(t) = 1$

$y(t) = e^{0.2t}$ $\qquad \dot{y}(t) = 0.2e^{0.2t}$

$$v(t) = \sqrt{(1)^2 + (0.2e^{0.2t})^2}$$

$$\theta(t) = \tan^{-1}\left(\frac{0.2e^{0.2t}}{1}\right)$$

$$w(t) = \frac{d\left(\tan^{-1}\left(0.2e^{0.2t}\right)\right)}{dt}$$

$dx \rightarrow 0.04e^{0.2t}$

$$w(t) = \frac{0.04e^{0.2t}}{1 + (0.2e^{0.2t})^2}$$

condiciones iniciales

$x(0) = 0$

$y(0) = e^{0.2(0)} = 1$

$\theta(0) = \tan^{-1}\left(0.2e^{0.2(0)}\right) = \tan^{-1}(0.2)$

Trayecto 2

cosenoidal

[8 18]

$f(x) = 3\cos(0.4(t-8))$

$x(t) = t$ $\qquad \dot{x}(t) = 1$

$y(t) = 3\cos(0.4(t-8))$ $\qquad \dot{y}(t) = -1.2\sin(0.4(t-8))$

$$v(t) = \sqrt{1 + (-1.2\sin(0.4(t-8))^2}$$

$$\theta(t) = \tan^{-1}(-1.2\sin(0.4(t-8))$$

$$w(t) = \frac{d\left(\tan^{-1}(-1.2\sin(0.4(t-8))\right)}{dt} \longrightarrow w(t) = \frac{-0.48\cos(0.4(t-8))}{1 + (-1.2\sin(0.4(t-8))^2}$$

$dx \rightarrow -0.48\cos(0.4(t-8))$

Trayecto 3

Logarítmica

[18 30]

$f(x) = 2\ln(t-17)$

$x(t) = t$ $\qquad$ $\dot{x}(t) = 1$

$y(t) = 2\ln(t-17)$ $\qquad$ $\dot{y}(t) = \dfrac{2}{t-17}$

$$v(t) = \sqrt{1 + \left(\frac{2}{t-17}\right)^2}$$

$$\theta(t) = \tan^{-1}\left(\frac{2}{t-17}\right)$$

$$\frac{-2}{(t-17)^2}$$

dx

$$w(t) = \frac{d\left(\tan^{-1}\left(\frac{2}{t-17}\right)\right)}{dt} = \frac{\frac{-2}{(t-17)^2}}{1 + \left(\frac{2}{t-17}\right)^2}$$